

Normalization & Activation Evolution

□□□□[Zhang & Sennrich, 2019] [Shazeer, 2020]

Normalization Schemes

Post-Norm

$x + \text{Sublayer} \rightarrow \text{LayerNorm} \rightarrow \text{Output}$



Pre-Norm

$\text{LayerNorm}(x) \rightarrow \text{Sublayer} \rightarrow x + \text{Output}$

- More stable training, no warmup needed
- Gradients flow through residual connections
- Deep models are trainable

All modern LLMs use Pre-Norm

Activation Functions

ReLU: $\max(0, x)$

Simple but has "dead neuron" problem

GELU: $x \cdot \Phi(x)$

Smooth transition, no dead neuron problem

SwiGLU: $(xV) \cdot \text{SiLU}(xW)$

Gating mechanism, perplexity reduced 0.5-1.0

□ Proven most effective

□ Gating enhances expressiveness

Used by LLaMA, Qwen, Mistral